A two-level biofilm model from one extended Hamilton principle

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Aim. In the thesis the growth field of Klempt et al. 2024 and the point model of Klempt et al. 2026 are solved separately and joined at the Gauss points. This needs choices that no paper fixes: how the void of the point model relates to the void of the field, whether the point model is scaled to the field, what drives the growth, and how the two clocks are linked. Here all equations follow from one energy, one dissipation potential and one constraint.

1 Why one level is not enough

The direct generalisation takes every species as a field $\phi_i(\mathbf{X},t)$ with one void $\phi_0 = 1-\sum_i \phi_i$ and the point-model energy at every point. The point model keeps every ϕ_i away from zero by its barrier energy $K/[\phi_i^2(1-\phi_i)^2]$, and its production is local. A 1D solution with all fields in one Newton system shows the consequence: the outer liquid fills up from the barrier floor everywhere at once, so the model cannot represent a biofilm next to liquid. The two voids are therefore different physical objects: the liquid outside the biofilm, and the liquid inside it. This motivates two levels.

2 State, energy, dissipation

$\phi(\mathbf{X},t) \in [0,1]$	biofilm phase (Klempt et al. 2024): biofilm against outer liquid
$s_i,\ s_0$	composition of the biofilm: species $i = 1 \dots n$ and inner liquid, $\sum_i s_i + s_0 = 1$
ψ_i	living fraction of species i (Klempt et al. 2026)
α	local expansion, $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$
$\phi s_i,\ \phi\sum_i s_i\psi_i$	absolute amount of species i, living biomass

With the weight $w = h(\phi)+\varepsilon$, $h(\phi) = \phi^2$ ($\varepsilon > 0$ small keeps the composition equations regular in the outer liquid, where they carry no weight), the energy density is

$$E = \frac{\beta}{2}|\nabla\phi|^2 + \lambda w \Psi_{\mathrm{pm}}(s, s_0, \psi) + w\frac{\beta_s}{2}\sum_i |\nabla s_i|^2 - k_\alpha(\alpha-1)\,\phi\sum_i s_i\psi_i + E_{\mathrm{mech}}, \tag{1}$$

with the point-model energy of Klempt et al. 2026

$$\Psi_{\mathrm{pm}} = K\Big[\sum_i B(s_i) + B(s_0) + \sum_i B(\psi_i)\Big] + \frac{\alpha^*}{2}\sum_i b_i\psi_i^2 - \frac{c^*}{2}\sum_{ij} A_{ij}\,s_i\psi_i\,s_j\psi_j, \qquad B(x) = \frac{1}{x^2(1-x)^2}, \tag{2}$$

and $E_{\mathrm{mech}} = \frac{\mu}{2}\phi^2(\mathbf{I}:\mathbf{C}_e - 3)$ as in Klempt et al. 2024 (dropped below for the same reason as there: its effect on growth is small). The dissipation potential is

$$D = \frac{\eta_\phi}{2}\dot\phi^2 + \lambda w D_{\mathrm{pm}} + \frac{\eta_\alpha}{2}\dot\alpha^2, \qquad D_{\mathrm{pm}} = \sum_i\Big[\frac{\eta}{2}\dot s_i^2 + \frac{\eta}{2}\big(\tfrac{\mathrm{d}}{\mathrm{d}t}(s_i\psi_i)\big)^2\Big] + \frac{1}{2}\dot s_0^2, \tag{3}$$

and the constraint is imposed with the multiplier field γ through $\lambda w\gamma(\sum_i s_i + s_0 - 1)$. The only new parameters are λ (the energy scale of the composition relative to the phase), β_s (spreading of the composition inside the biofilm) and the form of the growth energy.

The barrier terms. In the point model the barrier terms $KB(\cdot)$ only keep s_i, s_0, ψ_i inside $(0,1)$; they are a regularised indicator of these box constraints. In the one-level point model their value never matters, only their derivatives. Weighted with the biofilm amount, the value would act on the phase: at the calibration's start ($\psi_i = 0.999$) the barrier part of Ψ_{pm} is about 517, against about 8 for $\psi_i = 0.5$, so dying cells would make the biofilm grow. The barrier terms are therefore treated as a constraint: they carry the weight w held fixed, like the multiplier term, act on the composition only and do no work on ϕ. In (1), Ψ_{pm} then stands for its physical part (interactions and antibiotic) wherever it acts on ϕ.

The weight. With $h(\phi) = \phi$ the force of the composition on the phase, $\lambda h'(\phi)\Psi_{\mathrm{pm}} = \lambda\Psi_{\mathrm{pm}}$, does not vanish where there is no biofilm, and the outer liquid fills again (the same defect as the source $k_\alpha\alpha$ of Klempt et al. 2024 in the void). With $h = \phi^2$, the weight of the mechanical term of Klempt et al. 2024, $h'(0) = 0$ and the composition acts only where biofilm is.

3 Stationarity conditions

The extended Hamilton principle

$$\delta_q \int_\Omega E \,\mathrm{d}V + \int_\Omega \partial_{\dot q} D\, \delta q \,\mathrm{d}V = 0$$

for all variations δq of $q = \{\phi, s_i, s_0, \psi_i, \alpha, \gamma\}$, with $\mathbf{n}\cdot\nabla\phi = \mathbf{n}\cdot\nabla s_i = 0$ on the boundary, gives:

Composition (divided by $\lambda w > 0$):

$$KB'(s_i) + \gamma + \eta(1+\psi_i^2)\dot s_i + \eta s_i\psi_i\dot\psi_i - c^*\psi_i\Big(\sum_j A_{ij}s_j\psi_j + \frac{k_\alpha(\alpha-1)\phi}{\lambda w c^*}\Big) - \frac{\beta_s}{\lambda w}\nabla\cdot(w\nabla s_i) = 0, \tag{4}$$

$$KB'(s_0) + \gamma + \dot s_0 = 0, \tag{5}$$

$$KB'(\psi_i) + \alpha^* b_i\psi_i + \eta s_i\psi_i\dot s_i + \eta s_i^2\dot\psi_i - c^* s_i\Big(\sum_j A_{ij}s_j\psi_j + \frac{k_\alpha(\alpha-1)\phi}{\lambda w c^*}\Big) = 0, \tag{6}$$

$$\textstyle\sum_i s_i + s_0 - 1 = 0. \tag{7}$$

Without the growth term and without composition gradients these are exactly the point model of Klempt et al. 2026 at every point. The growth energy enters with the same structure as the interactions.

Phase:

$$\eta_\phi\dot\phi - \beta\nabla^2\phi + \lambda h'(\phi)\Psi_{\mathrm{pm}} + h'(\phi)\frac{\beta_s}{2}\sum_i|\nabla s_i|^2 - k_\alpha(\alpha-1)\sum_i s_i\psi_i = 0. \tag{8}$$

Here Ψ_{pm} is the physical part (interactions and antibiotic; the barrier terms act as a constraint, see above). The composition acts on the phase through it: where the composition energy is low (strong favourable interactions), the biofilm tends to grow, and the growth source is weighted by the living share. The dependence of the dissipation weight w on ϕ is not a force: the principle varies D with respect to the rates only.

Expansion:

$$\eta_\alpha\dot\alpha = k_\alpha\,\phi\sum_i s_i\psi_i. \tag{9}$$

4 Limits

1. $\lambda \to 0$, $\beta_s = 0$: the composition at every point is the point model on its own clock, started where the biofilm is, and (9) is the growth law of the element's scheme (c), $\dot\alpha = \kappa\,\phi_{3D}\sum_i\bar\phi_i$ with $\kappa = k_\alpha/\eta_\alpha$. Scheme (c) is thus the decoupled limit of this model, and its point model, which runs unscaled as in scheme (b), is the consistent one: its ϕ_0 is the inner liquid s_0.

2. With $\phi\sum_i s_i$ in place of the living biomass in the growth energy and one species, (9) is Eq. 36 of Klempt et al. 2024 up to the factor $\sum_i s_i$.

3. Scheme (a), which scales the point model to $\sum_i\phi_i = \phi_{3D}$, identifies the inner liquid with the outer one ($s_0 = 1-\phi$). It is not a limit of the model.

5 Energy

Proposition 1. *Let E not depend on time explicitly and let no external power act. Along solutions,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega E\,\mathrm{d}V = -2\int_\Omega D\,\mathrm{d}V \le 0.$$

Proof. Every stationarity condition has the form $\partial_q E + (\text{gradient terms}) + \partial_{\dot q} D + (\text{constraint terms}) = 0$. Multiply each by $\dot q$, integrate over Ω and add. The gradient terms integrate by parts to the rate of the gradient energies (the boundary terms vanish by the no-flux conditions); the constraint terms give $\lambda w\gamma\,\frac{\mathrm{d}}{\mathrm{d}t}(\sum_i s_i + s_0) = 0$; and D is quadratic in the rates, so $\sum_q \dot q\,\partial_{\dot q} D = 2D$. What remains is $\frac{\mathrm{d}}{\mathrm{d}t}\int E = -\int 2D$. □

Remark 1. *Switching an antibiotic on (α^* changing in time) adds $\int \partial_t E = \int \lambda w \frac{\dot\alpha^*}{2}\sum_i b_i \psi_i^2$, an external input; this is the jump seen at the switch in the 1D runs. Klempt et al. 2024's front term h_ϕ^* is not derived from a potential, so it is left out here; a model that includes it is no longer a gradient flow.*

6 Discretisation and checks

1D, nodes with mirrored ends, all unknowns of all nodes in one residual, implicit Euler, full Newton to $|R| < 10^{-10}$ with damping that keeps $s, \psi \in (0,1)$ and $\phi > 0$ (`JAXFEM/unified_two_level_1d.py`). The residual is not written by hand: it is the automatic derivative (JAX) of the discrete functional $E_h + \Delta t\, D_h + C_h$ of one step, so the discrete equations are its Euler–Lagrange equations by construction. With one node and no coupling the composition rows equal the point model as coded in this repository to round-off, so the point model is the stationarity condition of the energy and dissipation used here. For the one-level model the same scheme reproduces the converged point model to 1.6×10^{-13} and the discrete energy decreases at every step (`unified_1d_monolithic.py`).

7 First results (1D, seed at the substrate, CH calibration)

Seed $\phi = 0.9$ on $x \le 0.25$, outer liquid $\phi = 10^{-4}$; $\lambda = 1$, $k_\alpha = 2$, $\beta = \beta_s = 0.02$, $T^* = 1$, 60 steps (120 steps change α by 0.7 % of the growth).

	α at the substrate	ϕ at $x = 0,\ 0.3,\ 1$	living share at $x = 0$
no antibiotic	1.195	0.92, 0.32, 10^{-4}	0.58
antibiotic from $T^* = 0.5$ ($\alpha^* = 0.65$, $b_i = 1$)	1.171	0.78, 0.27, 10^{-4}	0.48

The outer liquid stays empty. The antibiotic lowers the growth by about 12 % and the biofilm phase itself (the composition acts on the phase), which the element's scheme (c) cannot do. The energy inequality of the proposition holds for the model with the barrier terms as stored energy; with the barrier terms as a constraint it holds in the limit $K \to 0$, and a discrete check is still to be done.

8 What remains assumed

The form of the growth energy (living biomass, and $\alpha - 1$ so that it vanishes before growth); the weight $h(\phi) = \phi^2$; the treatment of the barrier terms as a constraint; λ and β_s; the clock of the composition relative to the phase, which is now the ratio of the dissipation coefficients η/η_ϕ and can be fixed by the measured volume; the stiffness; detachment.